		$T = 800\ (2) = 1600$ N
4	**A**	Assuming the mass of each ball is m. Since total initial momentum = mv By Conservation of linear momentum, total final momentum should be mv A) Total final momentum = $0 + mv = mv$ B) Total final momentum = $\frac{mv}{2} + \frac{mv}{2} = mv$ C) Total final momentum = $\frac{3mv}{4} - \frac{mv}{4} = \frac{1}{2}mv$ D) Total final momentum = $mv - mv = 0$ Only A and B’s momentum are conserved. Elastic means relative velocity of approach = relative velocity of separation By relative velocity of approach and separation, KE not conserved for option B.
5	**A**	By conservation of energy, elastic potential energy is converted to kinetic energy and